

SEDAR 44 Atlantic Red Drum

Data/Assessment Workshops

This includes research recommendations developed by the Red Drum Technical Committee and Red Drum Stock Assessment Subcommittee for the SEDAR 44 Stock Assessment. Research recommendations are divided into short and long term projects. Short and long term research recommendations are prioritized, with the highest priorities listed first under each section and the lowest priorities listed last under each section.

Short Term

- Conduct experiments using logbooks to develop estimates of the B2 catch length composition in both the North and South regions.
- Determine if existing and historic recreational data sources (e.g., tagging) can be used to evaluate better B2 selectivities.
- Further study is needed to determine discard mortality estimates for the Atlantic coast, both for recreational and commercial gears. Additionally, discard estimates should examine the impact of slot-size limit management and explore regulatory discard impacts due to high-grading. Investigate covariates affecting discard mortality (e.g., depth, size, seasonality).
- Continued and expand observer coverage for the NC and VA gill net fisheries (5-10% coverage).
- Expand observer coverage to include other gears of concern (i.e. haul seine, pound net, trawls). Expand biostatistical sampling (ages and lengths) to better cover all statistical strata (gears/states - principally NC and VA) and collect more ages proportional to lengths, preferably otoliths. Conduct statistical analysis to determine appropriate sample sizes to adequately characterize the age-size composition of removals.
- Conduct a tagging study using emerging technologies (i.e., acoustic tagging, satellite tagging, genetic tags) to evaluate stock mixing and identify movement of sub-adult fish transitioning to maturity.
- Determine batch fecundity estimates of red drum. Need to include age-specific spawning frequency and spawning season length for this indeterminate spawner.
- Update maturity schedules for Atlantic red drum from Florida to Virginia. Preferably, gonad histology samples should be collected from all sizes over time and archived.
- Otolith microchemistry analysis should be considered to look at state level differences between regions to support stock structure differentiation.
- Continue cooperation between state ageing labs, such as the October 2008 red drum ageing workshop, to provide consistent age verification between labs.

Long Term

- Investigate iterative re-weighting of data components to identify the appropriate weights given to each data component in the objective function.
- Investigate alternative functions for retention to include recreational harvest and dead releases in the same fleets. Commercial discards should also be considered as a discard component of the landings fleet.

- Allow for time varying reporting rate of tag recaptures in the assessment model. This would allow use of more recent tag-recapture data from NC and estimates of changes over time in both regions.
- Continue genetic analyses (i.e, SC DNR analyses) to evaluate stock structure and mixing and temporal changes in genetic composition of the red drum population.
- Consider a pilot Virginia adult survey and expanding current adult fishery-independent survey coverage in Florida waters.
- Identify impacts of water quality, environmental, and ecosystem changes on red drum stock dynamics. Incorporate in the stock assessment models.
- Quantify habitat changes for future management planning

Review Workshop

Research recommendations were presented to the Panel in working document RW01 and are reiterated below in italic font followed by comments by the Panel. Generally speaking, the Panel agreed with the list of recommendations, as well as the prioritization. Additional recommendations by the Panel are included at the end of the list provided by the AT. The Panel noted that the recommendations generally fell into two categories: those that addressed gaps in life history information, and those that addressed issues with the model. While both are important, it was recommended that the research to address model concerns should be given higher priority.

Short Term

- *Conduct experiments using logbooks to develop estimates of the B2 catch length composition in both the North and South regions.*

Recreational releases are becoming an increasingly important component of the total catch. Unfortunately, the recreational survey does not collect length information from released fish, so length composition for this sector was characterized with length frequency data borrowed from available tagging data. The Panel noted several concerns surrounding the tagging data, such as minimum size requirements for tagging. The Panel therefore agrees that this research recommendation receive a high priority in the short term. For the long term, the Panel recommends modeling the recreational harvest and discards as a single fleet with a discard function; however, this will require assistance from SS3 developers since SS3 software currently only allows a logistic retention function that is not appropriate for a slot limit. It was cautioned that the combined fleet should exclude any fleet that has “non-standard” discarding practices (e.g., releases all red drum regardless of size) that might have a different selectivity pattern.

- *Determine if existing and historic recreational data sources (e.g., tagging) can be used to evaluate better B2 selectivities.*

The Panel noted that the selectivity pattern of recreational live releases is expected to be bimodal and cannot be modeled with the double normal selectivity option used for the other fleets. The Panel recommended investigating the non-parametric selectivity function available in SS3 in the short term, and modelling recreational live releases with recreational harvest as a single fleet in the long term (see previous recommendation).

- *Further study is needed to determine discard mortality estimates for the Atlantic coast, both for recreational and commercial gears. Additionally, discard estimates should examine the impact of slot-size limit management and explore regulatory discard impacts due to high-grading. Investigate covariates affecting discard mortality (e.g., depth, size, seasonality).*

A better understanding of discard mortality rates in the commercial and recreational sectors, and the covariates that influence mortality rates, is important to adequately characterize the removals by these sectors. This and the previous two recommendations should receive high priority to gain a better understanding of discard practices which are becoming a larger component of the fishery.

- *Continued and expand observer coverage for the NC and VA gill net fisheries (5-10% coverage).*

It was noted that the scale of discards in the gill net fishery is substantial, and an accurate characterization of these fish should be included in the total removals. Priority for this research/monitoring should be based on the scale of commercial discard removals relative to the recreational release mortalities.

- *Expand observer coverage to include other gears of concern (i.e. haul seine, pound net, trawls).*

The AT confirmed that harvest from these gears is substantially less than gill nets, and discard mortality is expected to be low. The Panel recommended that priority of this research recommendation be evaluated relative to other sectors of the red drum fishery in terms of their overall contribution to removals.

- *Expand biostatistical sampling (ages and lengths) to better cover all statistical strata (gears/states - principally NC and VA) and collect more ages proportional to lengths, preferably otoliths. Conduct statistical analysis to determine appropriate sample sizes to adequately characterize the age-size composition of removals.*

The Panel commented that having sufficient data is certainly important to inform the model, but cautioned that sampling for the sake of increasing sample size can be counterproductive. Targeted sampling plans should be developed that fill identified data gaps and improve the model and/or management decisions but minimize over sampling. It may be necessary to collect large numbers of samples in the short term to better understand key life history characteristics, such as growth patterns, but over time sample sizes could be scaled back to a maintenance level of sampling.

- *Conduct a tagging study using emerging technologies (i.e., acoustic tagging, satellite tagging, genetic tags) to evaluate stock mixing and identify movement of sub-adult fish transitioning to maturity.*

The Panel noted that the information presented on stock structure was sufficient and informative, so while a tagging program could provide useful information, it is more of a long term issue. The AT expressed a concern that there may be substantial mixing of stocks in offshore waters. However, harvest from the offshore areas is low, so there should be little concern for mixed stock harvest. The Panel suggested that this research recommendation receive a lower priority and also be considered as a long term monitoring project along with otolith microchemistry and genetic analyses (discussed below).

- *Determine batch fecundity estimates of red drum. Need to include age-specific spawning frequency and spawning season length for this indeterminate spawner.*

The model does not currently use fecundity information, and using spawning stock biomass as a proxy for fecundity is a commonly accepted practice, so this does not need to be considered a high priority research recommendation. That being said, basic life history information such as fecundity is important to have, particularly as it relates to size and age. For example, it is known that fecundity of some sciaenids is not linear with spawner biomass, and therefore the metric used may have management implications.

- *Update maturity schedules for Atlantic red drum from Florida to Virginia. Preferably, gonad histology samples should be collected from all sizes over time and archived.*

The Panel agreed that collecting histological samples to standardize maturity classification, confirm maturity schedules, and provide a reference collection is important but cautioned that such a program may be expensive. Cost should therefore be considered during prioritization of sample collection.

- *Otolith microchemistry analysis should be considered to look at state level differences between regions to support stock structure differentiation.*

As with the research recommendation on archival tagging, the Panel commented that the stock structure information provided during the RW was sufficient and informative. It was recommended that this research be considered as a long term project in conjunction with the genetic analysis (discussed below).

- *Continue cooperation between state ageing labs, such as the October 2008 red drum ageing workshop, to provide consistent age verification between labs.*

The Panel noted that coordination/verification of ageing practices is very important and that work should be ongoing and conducted at regular intervals. The Panel also recommended that the work be broadened to include coordinating the collection of age samples and the development of an ageing error matrix.

Long Term

- *Investigate iterative re-weighting of data components to identify the appropriate weights given to each data component in the objective function.*

The Panel recommended that this work be viewed as a short term recommendation in terms of improving the model. In addition, the Panel noted that the work should include additional methods for weighting the model inputs, with iterative reweighting as one possible method.

- *Investigate alternative functions for retention to include recreational harvest and dead releases in the same fleets. Commercial discards should also be considered as a discard component of the landings fleet.*

The Panel agreed that combining the recreational harvest and discards into a single fleet would be more appropriate but reiterated that SS3 does not currently have that functionality given the characteristics of the red drum fishery (slot limit). In addition, this step is not necessary to get a solid working model, so the Panel recommended that this work receive lower priority.

- *Allow for time varying reporting rate of tag recaptures in the assessment model. This would allow use of more recent tag-recapture data from NC and estimates of changes over time in both regions.*

The Panel recommended that both regions should continue their tagging programs to evaluate tag return rates, but identified two concerns with this research recommendation. First, given the available data, it is unlikely that SS3 would be able to reliably estimate time varying reporting rates. Second, even if the functionality were available, the data should be evaluated prior to use in the model to confirm that they are informative. It was cautioned that including uninformative data on this (or any other) parameter may be counterproductive to the model fitting process.

- *Continue genetic analyses (i.e., SC DNR analyses) to evaluate stock structure and mixing and temporal changes in genetic composition of the red drum population.*

Details regarding stock structure and stock mixing rates are important to understand in non-migratory stocks that extend over wide geographic scales. It was noted that the information pertaining to Atlantic red drum stock structure was informative, with the split between northern and southern, and southern and Gulf stocks clearly defined. However, monitoring of mixing rates and confirmation of stock boundaries is important over the long term, as well as investigating the possibility for finer scale stock structure, as new techniques are developed. Several research recommendations addressed stock structure and mixing, including acoustic tagging, otolith microchemistry, and genetic analysis. It was recommended that long term monitoring be conducted at modest levels of sampling using a combination of these three techniques.

- *Consider a pilot Virginia adult survey and expanding current adult fishery independent survey coverage in Florida waters.*

The Panel commented that fishery independent sampling should be representative of the entire population. The AT should evaluate the adequacy of current sampling levels and expand as necessary.

- *Identify impacts of water quality, environmental, and ecosystem changes on red drum stock dynamics. Incorporate in the stock assessment models.*

The Panel agreed that understanding external drivers on stock dynamics is important, but noted that this research recommendation is very broad and generic. Preliminary work should be done using available data to identify potential factors that affect red drum so that more directed work could be conducted moving forward. It was also suggested that the work be conducted by non-assessment biologists since it is less of a priority.

- *Quantify habitat changes for future management planning*

The Panel commented that the specific components of the previous recommendation could all be considered aspects of habitat. In that regard, this recommendation which addresses physical habitat can be included in with the previous recommendation.

Additional research recommendations identified by the Panel

In addition to the short and long term research recommendations provided by the Red Drum Technical Committee, the Panel identified a number of items that could improve the current model and future assessments.

The AT should conduct a comprehensive review of tag reporting rates, including identifying additional data sources, evaluating alternative estimation methodologies, and/or using information from similar species. This work should be done in the short term to update (or corroborate) data currently being used. The final value used should be sufficiently justified because of its apparent impact on the scale of model results.

As discussed under TOR 3, the AT should investigate the treatment of age and length data to fit selectivities for the longline survey. During this exploration the team should pay particular attention to the fitting of the growth function and its effect on model fitting for the other surveys, age composition, and historic recruitment; changes in selectivity estimates for the different fleets and fits to other datasets should also be examined for possible indication of conflict in the signals provided.

The Panel noticed an abnormal length frequency distribution (certain lengths in the heart of the distribution with 0 frequency) in the northern GNBS data which is likely an artifact of converting lengths in inch bins to centimeter bins. These data could be influencing the length composition fits (and therefore overall model results) and should be rectified before moving forward.

The AT should explore the effect of changing sample size cutoffs, such as the number of tag returns necessary for each tag group (currently using $N = 300$) or minimum sample size before data borrowing is necessary. This work should be viewed as a diagnostic and sensitivity exercise and should be explored in the short term but only after a solid base model is available.

One potential avenue for exploration could be to input size at age directly based on external estimates of growth. This would allow for non von Bertalanffy growth model to see if a different growth model addresses some of these perceived issues.

For this and future assessments, the Panel reiterates their recommendation to start simple and add complexity only after the model is providing credible results and the AT understands how the model is performing. This is particularly important when new data sources or alternative data treatments are being considered. Once a “simple” model has been developed, the Panel recommends investigating two “add-ons” independently of each other (e.g. extending the time series vs. adding selectivity blocks), and selecting the one that provides more information as the next scenario. This process should be repeated, adding complexity in a step-wise fashion, until sufficient complexity is achieved or model performance breaks down.

As noted in TOR 1, the Panel recommends additional time be spent evaluating available data sources prior to their use in the model. Simple exercises such as checking whether the different data sources

corroborate one another or indices can track strong year classes can help determine the information content of a particular data source. A thorough understanding of the input data strengths and weaknesses can be useful to pinpoint issues in model performance.